The commutator structure of canonical transcorrelation

This is the skeleton of the CT derivation as it is written on the sheets: the ansatz, the second-order expansion, and the decomposition of the commutators that has to be carried out. Section 4 lists every block the derivation needs, with the rule that produces it and its present status. Nothing here is a new result; the point is to fix the structure once, so the block-by-block algebra of `progress_note/singles_derivation_plan.md` has a single home.

1. The ansatz

$$\hat{H} = \hat{F} + \hat{V}, \qquad \hat{F} = f^q_p \, \hat{E}^q_p, \qquad \hat{V} = \tfrac{1}{2} \, g^{rs}_{pq} \, \hat{E}^{pq}_{rs}, \tag{1}$$

$$\hat{A} = \hat{T} - \hat{T}^\dagger, \qquad \hat{T} = \hat{T}_1 + \hat{T}_2, \qquad \hat{T}_1 = t^a_i \, \hat{E}^i_a, \qquad \hat{T}_2 = \tfrac{1}{2} \, t^{ab}_{ij} \, \hat{E}^{ij}_{ab}. \tag{2}$$

The transformation is

$$\hat{\bar{H}} = e^{-\hat{A}} \, \hat{H} \, e^{\hat{A}} = e^{\hat{A}^\dagger} \hat{H} \, e^{\hat{A}}, \tag{3}$$

the two forms being the same because A is anti-Hermitian, $\hat{A}^\dagger = -\hat{A}$. Expanding and keeping terms through second order,

$$\hat{\bar{H}} = \hat{H} + [\hat{H}, \hat{A}] + \tfrac{1}{2} [[\hat{F}, \hat{A}], \hat{A}] + \cdots , \tag{4}$$

where the Fock operator replaces $\hat{H}$ in the double commutator — the second-order truncation, not an identity. In the F12 case this is Yanai and Shiozaki's own truncation [2],

$$\hat{\bar{H}}^{F_{12}} \approx \hat{H} + [\hat{H}, \hat{A}_{F_{12}}]_{1,2} + \tfrac{1}{2} \big[[\hat{F}, \hat{A}_{F_{12}}]_{1,2}, \hat{A}_{F_{12}} \big]_{1,2},$$

and the same form with the explicit single excitation added is Kumar *et al.* [3], eqs. (3)–(4). Two conventions have to be stated with it:

- **The $[\dots]_{1,2}$ subscript.** Only one- and two-body operators generated by a commutator are kept; the three-body pieces are folded back with the extended normal ordering (cumulants) of Kutzelnigg and Mukherjee [1]. This is an approximation and is recorded as one.
- **The single excitation.** $\hat{T}_1$ is Kumar's addition; Yanai and Shiozaki's $\hat{A}^{F12}$ contains the projected geminal only. Everything below is written with $\hat{T}_1$ present and reduces to the geminal-only case at $t^a_i = 0$.

2. The decomposition

Because $\hat{A} = \hat{T} - \hat{T}^\dagger$ and $\hat{H}$ is Hermitian,

$$[\hat{H}, \hat{A}] = [\hat{H}, \hat{T}] + [\hat{H}, \hat{T}]^\dagger, \qquad \big([\hat{H}, \hat{T}]\big)^\dagger = -[\hat{H}, \hat{T}^\dagger], \tag{5}$$

so only the $\hat{T}$ half of every commutator has to be computed. Splitting $\hat{H} = \hat{F} + \hat{V}$ and $\hat{T} = \hat{T}_1 + \hat{T}_2$,

$$[\hat{H}, \hat{T}] = [\hat{F}, \hat{T}_1] + [\hat{F}, \hat{T}_2] + [\hat{V}, \hat{T}_1] + [\hat{V}, \hat{T}_2], \tag{6}$$

$$[\hat{F}, \hat{A}] = \hat{C} = \hat{C}_1 + \hat{C}_2, \qquad \hat{C}_i = [\hat{F}, \hat{A}_i] = [\hat{F}, \hat{T}_i] + [\hat{F}, \hat{T}_i]^\dagger. \tag{7}$$

The double commutator collapses before it is expanded. Since $\hat{A}$ is anti-Hermitian, $\hat{C} = [\hat{F}, \hat{A}]$ is Hermitian, so $[\hat{C}, \hat{T}]^\dagger = -[\hat{C}, \hat{T}^\dagger]$ and, using (7),

$$\tfrac{1}{2}[[\hat{F}, \hat{A}], \hat{A}] = \tfrac{1}{2}[\hat{C}, \hat{T}] + \text{h.c.} = \tfrac{1}{2}[\hat{C}_1, \hat{T}] + \tfrac{1}{2}[\hat{C}_2, \hat{T}] + \text{h.c.}. \tag{8}$$

Expanding the outer commutator in the two $\hat{T}$'s,

$$\tfrac{1}{2}[[\hat{F}, \hat{A}], \hat{A}] = \tfrac{1}{2}\Big([\hat{C}_1, \hat{T}_1] + [\hat{C}_1, \hat{T}_2] + [\hat{C}_2, \hat{T}_1] + [\hat{C}_2, \hat{T}_2]\Big) + \text{h.c.}, \tag{9}$$

which is the statement of the sheets at steps ⑤ and ⑥: four double commutators, eight commutators in the $\hat{T}$'s.

Two remarks on (8). The inner commutator is with $\hat{F}$ — it is what produces $\hat{C}$ — and only the *outer* one is with $\hat{T}$; writing the middle step with $\hat{A}$ inside, $\frac{1}{2}[[\hat{C}_i, \hat{A}], \hat{T}]$, gives a different operator, because $[[\hat{C}_i, \hat{A}], \hat{T}] = [[\hat{C}_i, \hat{T}], \hat{T}] - [[\hat{C}_i, \hat{T}^\dagger], \hat{T}]$. And the conjugate half is not an independent computation: (8) says the whole double commutator is $[\hat{C}, \hat{T}]$ plus its own conjugate, so only four of the eight commutators of (9) have to be evaluated. Both readings are checked on explicit matrices in `tests/test_ct_structure.py`: (8) holds to machine precision, and the reading with $\hat{A}$ inside is off by a matrix norm of order 10^4.

The four blocks are *not* pairwise equal —

$$\hat{T}_1\hat{T}_2 - \hat{T}_2\hat{T}_1 \neq 0 \tag{10}$$

(4 terms), so the Jacobi shortcut $[[\hat{F}, \hat{T}_1], \hat{T}_2] = [[\hat{F}, \hat{T}_2], \hat{T}_1]$ is invalid and the difference $-[[\hat{T}_1, \hat{T}_2], \hat{F}]$ has to be carried. Any place that used that shortcut has to be redone.

3. The rules the blocks are built from

Every block in §4 is a sum of products of two $\hat{E}$ operators, so the whole derivation needs only the four product rules and the four commutators of the $\hat{E}$ operators, at ranks $(1,1)$, $(1,2)$, $(2,1)$ and $(2,2)$. In the paired order — every join coefficient $+1$, the residual's blocks listed re-paired first, then the left leftovers, then the right leftovers — they are

$$[\hat{E}^q_p, \hat{E}^s_r] = \delta^s_p\, \hat{E}^q_r - \delta^q_r\, \hat{E}^s_p, \tag{11}$$

$$[\hat{E}^q_p, \hat{E}^{tu}_{rs}] = \delta^t_p\, \hat{E}^{qu}_{rs} + \delta^u_p\, \hat{E}^{qt}_{sr} - \delta^q_r\, \hat{E}^{tu}_{ps} - \delta^q_s\, \hat{E}^{ut}_{pr}, \tag{12}$$

$$[\hat{E}^{tu}_{rs}, \hat{E}^q_p] = -[\hat{E}^q_p, \hat{E}^{tu}_{rs}], \tag{13}$$

and the $(2,2)$ commutator, whose four-body part cancels and which leaves eight three-body and four two-body terms,

$$\begin{aligned}[\hat{E}^{rs}_{pq}, \hat{E}^{uw}_{tv}] = {}& \delta^u_p\, \hat{E}^{rsw}_{tqv} + \delta^w_p\, \hat{E}^{rsu}_{vqt} + \delta^u_q\, \hat{E}^{srw}_{tpv} + \delta^w_q\, \hat{E}^{sru}_{vpt} \\ & + \delta^u_p\delta^w_q\, \hat{E}^{rs}_{tv} + \delta^w_p\delta^u_q\, \hat{E}^{rs}_{vt} \\ & - \delta^r_t\, \hat{E}^{uws}_{pvq} - \delta^s_t\, \hat{E}^{uwr}_{qvp} - \delta^r_v\, \hat{E}^{wus}_{ptq} - \delta^s_v\, \hat{E}^{wur}_{qtp} \\ & - \delta^r_t\delta^s_v\, \hat{E}^{uw}_{pq} - \delta^s_t\delta^r_v\, \hat{E}^{uw}_{qp}. \end{aligned} \tag{14}$$

All four are verified against explicit Jordan–Wigner matrices over every label assignment, in the spin-orbital *and* the spin-free Fock space, by `check_paired_commutators.py` in the singles directory: 0 failing assignments, worst deviation 0.0.

4. The blocks

block	rule used	rank out	terms	status
$[\hat{F}, \hat{T}_1]$	(11)	1	2	verified
$[\hat{F}, \hat{T}_2]$	(12)	2	4	verified
$[\hat{V}, \hat{T}_1]$	(12)	2	4	verified
$[\hat{V}, \hat{T}_2]$	(14)	2 (+3)	10	*blocked*
$[\hat{T}_1, \hat{T}_2]$	(12)	2	4	verified
$[\hat{F}, \hat{A}_1]$	(11), (5)	1	2	verified
$[\hat{F}, \hat{A}_2]$	(12), (5)	2	4	verified
$\frac{1}{2}[[\hat{F}, \hat{A}_1], \hat{T}_1]$	(11)	≤ 2	2	verified
$\frac{1}{2}[[\hat{F}, \hat{A}_1], \hat{T}_2]$	(12)	≤ 2	8	verified
$\frac{1}{2}[[\hat{F}, \hat{A}_2], \hat{T}_1]$	(12)	≤ 2	8	*blocked*
$\frac{1}{2}[[\hat{F}, \hat{A}_2], \hat{T}_2]$	(14)	≤ 2	80	*blocked*

The three blocked blocks are blocked by one bug, not by the algebra. The blocks that fail their numerical check are exactly those whose expansion needs the $(2, 2)$ rule (14) — which is itself verified. The fault is in the canonicalisation step of the symbolic engine (`ctformula/algebra.py`, `canon_op`): it brings an operator to canonical form by sorting the upper and the lower index row *independently* and taking the sign of the two sorts. For a spin-orbital monomial that is correct — the monomial is fixed by the two multisets, up to sign. For a spin-free $\hat{E}$ it is not: sorting the two rows independently changes the *pairing* of upper to lower, which is part of the operator, so terms that differ only by such a change are wrongly merged and their coefficients inherited from a relisting sign that the spin-free operator does not have. The engine's own spin-orbital checks pass, and its spin-free ones fail, on exactly the same expressions — which is the signature of this bug.

The fix is to canonicalise a spin-free $\hat{E}$ as a set of pairs (u_i, l_i) with the sign of the pair permutation, keeping the row-wise canonical form for the spin-orbital monomials; the convention must be carried through the engine, because it cannot be read off the operator alone. Until that is done, the three blocked blocks must not be quoted: the term lists they produce now carry the wrong pairing on some terms, and the failing numeric sweeps are what exposes it.

5. What is verified, and what is not

- **Verified.** The four rules of §3, in both conventions, over every label assignment; and the eight blocks marked verified in §4, by a Fock-space sweep and an independent symbolic re-derivation, which is what `tests/test_ct.py` reports (3 of its 4 checks pass).
- **Blocked.** $[\hat{V}, \hat{T}_2]$, $\frac{1}{2}[[\hat{F}, \hat{A}_2], \hat{T}_1]$ and $\frac{1}{2}[[\hat{F}, \hat{A}_2], \hat{T}_2]$ — and with them the explicit $\bar{h}$ and $\bar{g}$, which cannot be collected until they are right.
- **Open beyond that.** The fold of the three-body pieces back into one- and two-body form ($[\dots]_{1,2}$), which is an approximation and has to be stated as one wherever the result is used; and the multireference generalisation, which needs the density and cumulant terms of the general reference rather than the vacuum contractions used here.

References

[1] W. Kutzelnigg and D. Mukherjee, *Normal order and extended Wick theorem for a multiconfiguration reference wave function*, J. Chem. Phys. **107**, 432 (1997).

[2] T. Yanai and T. Shiozaki, *Canonical transcorrelated theory with projected Slater-type geminals*, J. Chem. Phys. **136**, 084107 (2012).

[3] A. Kumar, A. Asthana, C. Masteran, E. F. Valeev, Y. Zhang, L. Cincio, S. Tretiak and P. A. Dub, *Quantum simulation of molecular electronic states with a transcorrelated Hamiltonian*, J. Chem. Theory Comput. **18**, 5312 (2022).